

Demonstration of Proposed Features

Date	1 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrate d (Yes / No)	Remarks
1	Dataset Analysis	Load and analyze the historical flood dataset using descriptive statistics and visualizations.	Implemented	Yes	Dataset successfully prepared and analyzed using Pandas and NumPy.
2	Data Preprocessing	Split the dataset, apply feature scaling using StandardScaler, and prepare data for model training.	Implemented	Yes	Preprocessing completed using Scikit-learn with StandardScaler.
3	Machine Learning Model	Train and evaluate Decision Tree, Random Forest, KNN, and XGBoost models and automatically select the best-performing model.	Implemented	Yes	Multiple models evaluated; the best model was selected based on the highest F1-score.
4	Flood Prediction System	Predict flood probability and classify the result into Low, Moderate, High, or Severe risk levels using user-provided weather parameters.	Implemented	Yes	Prediction performed using the saved trained model and preprocessing scalar.
5	Flask Web Application	Develop a web interface for entering inputs and displaying prediction results.	Implemented	Yes	Home, Prediction, About, and Result pages developed successfully using Flask, HTML, CSS.

Feature Implementation Summary:

Total Features Proposed	5
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	100%